

New approaches to fish aging for stock assessments Introduction

Aaron Adamack
DFO – NL Region



Territorial Acknowledgement



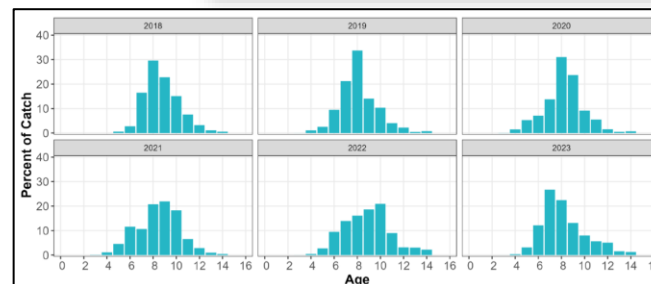
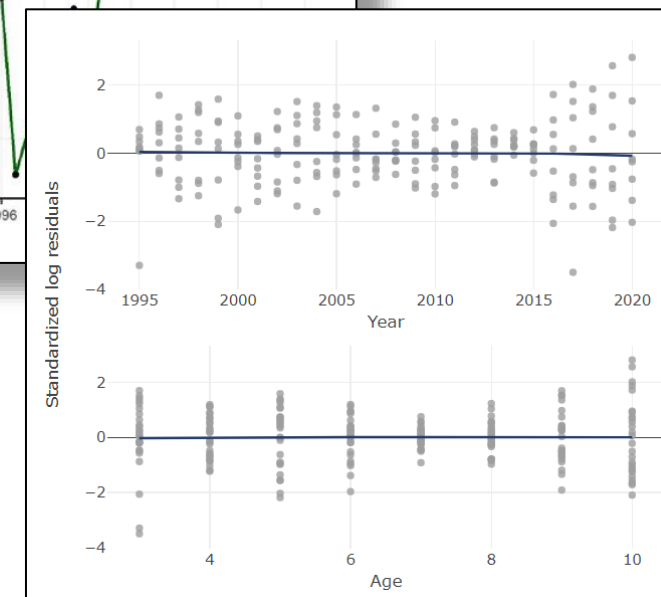
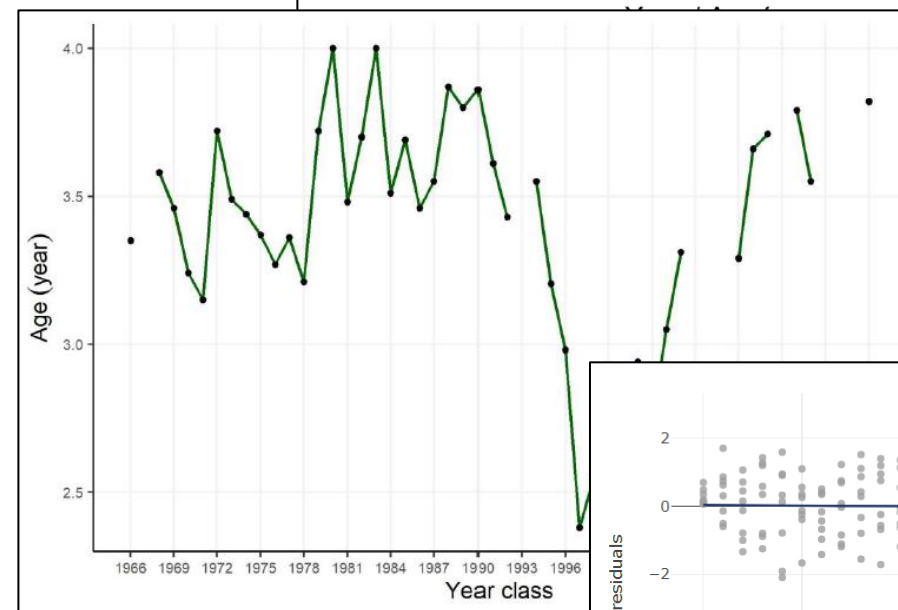
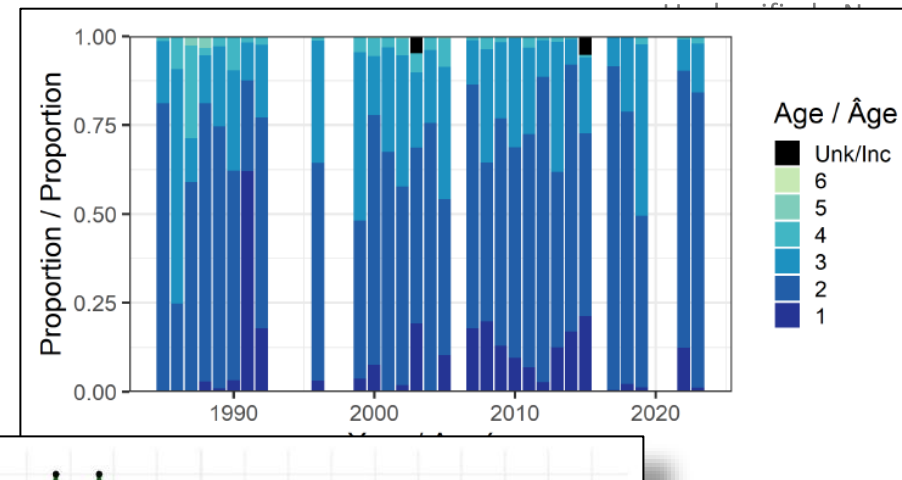
https://youtu.be/zrk7ljagNRQ?si=XuJ2ee8A_m97N3OC

Introduction

- Aging hard parts is an important, but costly part of stock assessments for fisheries agencies worldwide
- Age-based population models are often considered the “gold standard” in stock assessments (Ricard et al. 2024)
- Fisheries and Oceans Canada ages 100k's of otoliths/yr across multiple programs costing \$\$\$\$\$\$
 - CSIRO estimated the cost at ~\$30AUD/otolith in 2015

Introduction

- Ages show up in all sorts of forms in our stock assessments.
- They are a key component of many stock assessments
- Needed for calculating key biological rates such as age of maturity, and growth and mortality rates
- Also used for determining length and weights at age



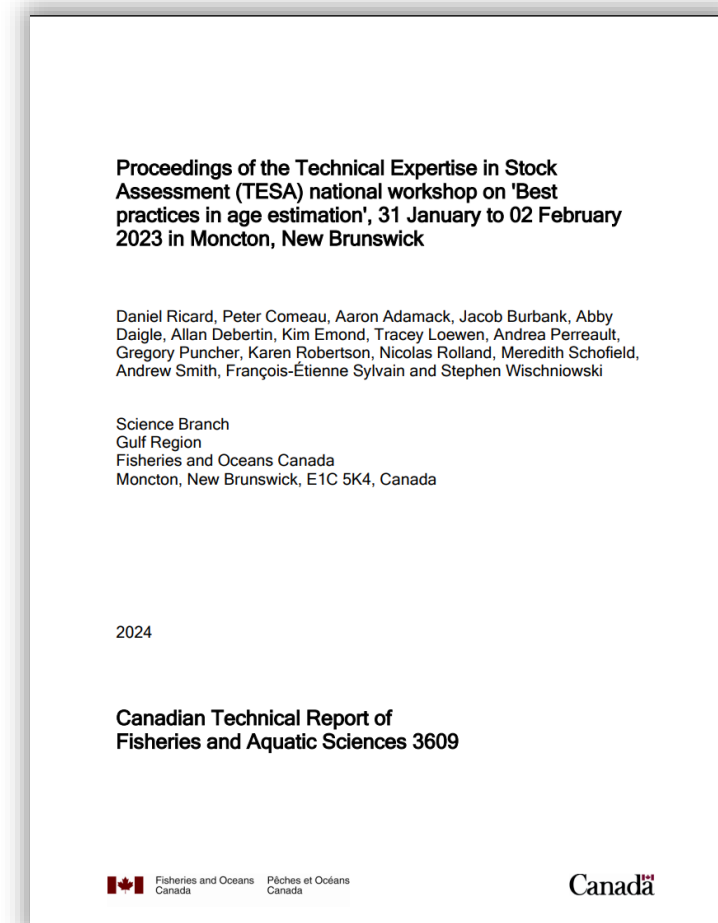
Introduction

- Demand for aging data has been growing as stock assessments have become increasingly quantitative in recent years (e.g. C68)
- There have been difficulties maintaining aging capabilities
- Losing a skilled technician can be very costly for an aging program
- Historically, fish aging was considered an entry-level position (e.g. “you’re just counting rings”); however, it’s quite skilled work taking up to 5 years to fully train people for multispecies labs (Ricard et al. 2024)
- In some cases, aging labs in Canada and the U.S. are reaching capacity

How do we deal with all of this?

National workshop on 'Best practices in age estimation'

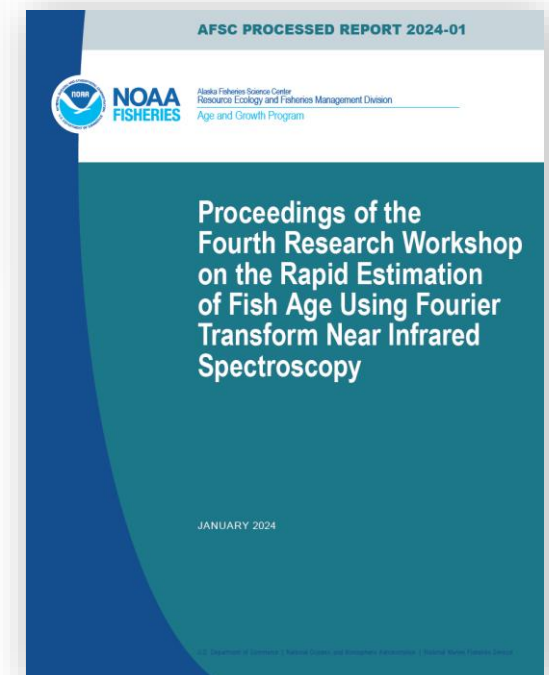
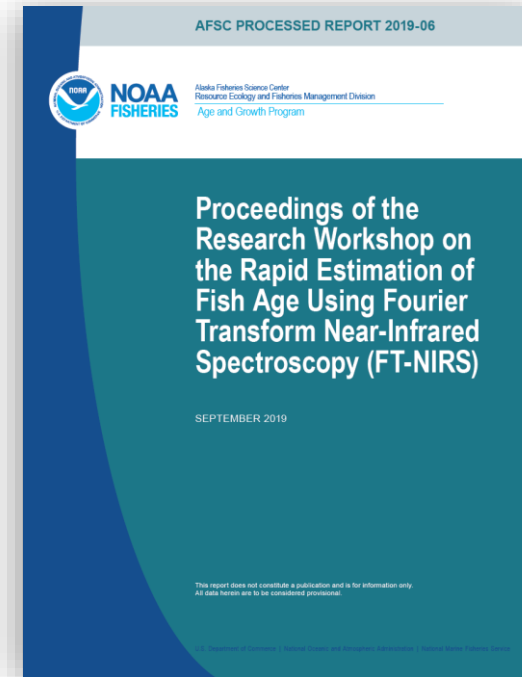
- In 2023, Daniel Ricard organized a TESA workshop on best practices in fish aging
- Provided an opportunity for DFO scientists involved in age estimation and age-based assessments, to share ideas and develop better methods
- Outcomes from the workshop included
 - Development of a set of best practices
 - Aging structure cataloguing, storage, and inventory
 - Development of reference collections
 - Laboratory operations
 - Training
 - Digital imaging of aging structures
 - Analysis of age estimates
 - Research and development
 - Recommended fostering a community of DFO scientists whose mandated tasks include <aging>



https://publications.gc.ca/collections/collection_2024/mpo-dfo/Fs97-6-3609-eng.pdf

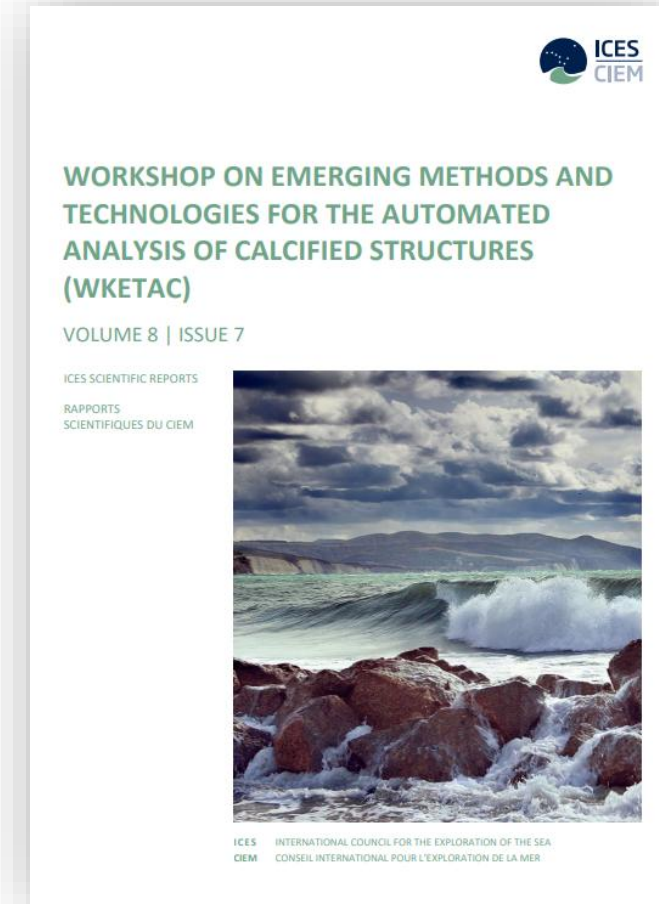
Speeding up age estimates using FT-NIR

- In the US, NOAA began investigating FT-NIR aging which was initially developed in Australia
- NOAA funded a 5-year Strategic Initiative in support of research and development of FT-NIR spectroscopy for fish aging in 2019
- (At least) Two really interesting proceedings came from this work
 - <https://repository.library.noaa.gov/view/noaa/21458>
 - <https://repository.library.noaa.gov/view/noaa/56377>



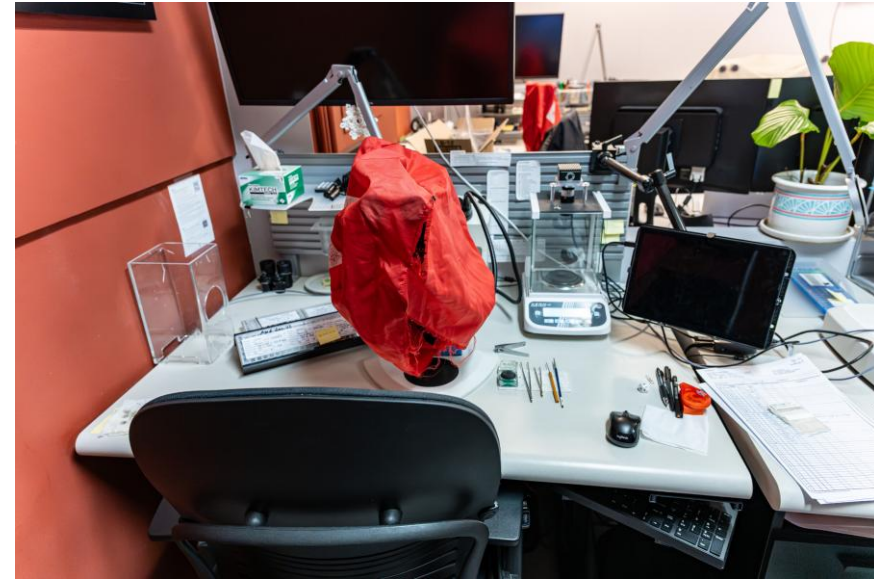
Other new methods in aging

- ICES held a workshop in October 2025 on Emerging methods and technologies for the automated analysis of calcified structures (WKETAC)
 - Organized by Côme Denechaud and Kélig Mahé
- Included participants from 20+ countries including Canada
- Had a number of presentations on the use of AI and FT-NIR for the aging of otoliths with a lot of focus on automation and standardization
- Also had extensive coverage of the use of 2D and 3D image analysis for stock identification
- Produced a set of target species and future pathways for developing a collaborative framework. Relevant species for Canada include
 - Atlantic cod
 - Atlantic salmon
 - Small pelagics (herring, mackerel and capelin)



How do we deal with all of this?

- A common message across these different meetings and working groups is that we need to collaborate more across the aging community
- Additionally, there is a lot of work happening today that is working towards automating and standardizing fish aging
- While there is a lot of interesting stuff happening in the field, it can often be unclear on who to speak to/work with
- I'm hopeful that this workshop will let people see some of the interesting work that is happening both in the department and internationally, develop new ideas for their own stocks, and perhaps help to initiate new collaborations both within the department and across institutions



Workshop structure

- As you'll have seen in the agenda, the two half days of the workshop will consist of a series of talks with a short break in the middle.
- I've left a lot of time around each presentation in order to give meeting participants lots of time to ask questions.
- If you're seeing something that could be useful for your own stocks, this is your opportunity to ask and perhaps get guidance on what you need to do to get started.
- If you think of something later or perhaps have a question that goes across multiple presentations, there'll be time at the end of each day for follow-up questions
- A talk by Karolina Jonko-Sobuś has been added to the agenda
 - *The application of IR spectroscopy in fish aging: pros and cons of the method*
 - Will be either end of today or second last talk on Wednesday